India AQI Dashboard: Working Prototype

1. Overview

The India AQI Dashboard is a fully functional, real-time platform designed to monitor air quality across India while pinpointing the exact industrial sources contributing to the pollution. It transitions the conversation from 'knowing the air is bad' to 'taking data-driven action.'

2. Core Features Demonstrated in the Prototype

- Real-Time Data Integration: Pulls live PM2.5 readings from 500+ OpenAQ monitoring stations.
- Interactive Mapping: Visualizes AQI data on a 5x5km geographic grid.
- Industrial Tracking: Maps nearly 4,000 facilities from Climate TRACE and Global Energy Monitor datasets.
- AI-Powered Action: Integrates Google Gemini to automatically draft context-aware legal complaint emails based on the real-time data displayed on the user's screen.
- Secure Authentication: Implements user login and session management backed by Neon PostgreSQL.

3. How the Prototype Works (System Flow)

- Data Ingestion: Background scripts fetch live data and process heavy datasets into lightweight runtime JSONs to ensure lightning-fast UI loads.
- Dashboard UI: Built with Streamlit, the interface provides a search engine, sorting, and color-coded status tiles based on the official India NAQI scale.
- Analytics Engine: When a user selects a city, the prototype instantly computes a 50x50km bounding box, assigns stations to grids, and calculates emission breakdowns by source (Industry, Transport, Crop Fires).
- AI Generation: The AI module reads the selected city's pollution metrics and factory leaderboard to construct a precise, actionable complaint letter, logging the action in the database for auditing.

4. Technical Stack

- Frontend: Streamlit, Folium, Altair
- Backend: Python, Pandas, Requests
- Database: Neon Serverless PostgreSQL
- AI & NLP: Google Gemini via LangChain